

SAFETY DATA SHEET

Nitrogen, Compressed Gas

SECTION 1: PRODUCT AND COMPANY IDENTIFICATION

1.1 Product Details

Product Name : Nitrogen, Compressed gas
Trade name : Industrial Nitrogen, Oxygen Free Nitrogen, Pure Nitrogen, high purity Nitrogen, Ultra High Purity Nitrogen
Identified uses : General industrial.

1.2 Company Identification

Company's Name : Leeden Gases Sdn. Bhd
Company's Address : Lot PT 5074,5075, Jalan Jangur 28/43,
Seksyen 28, 40400 Shah Alam, Selangor.
Emergency telephone Number : 03-55228222 (Hunting Line) / 03-55228288 (Hotline)

SECTION 2: HAZARD IDENTIFICATION

Classification according to Regulation (EC) No 1272/2008 as amended:

Gases under pressure - Compressed gas

Hazard pictograms :



Signal word : Warning
Hazard statements : H280 Contains gas under pressure; may explode if heated.
May displace oxygen and cause rapid suffocation
Simple Asphyxiant

Precautionary statements:

Storage : P403 Store in a well-ventilated place.
P410 Protect from sunlight
Environmental Effects : Not harmful.
Other Hazards : The rapid release of compressed gas may cause frostbite

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Chemical name : Nitrogen
CAS number : 7727-37-9

Proportion : >99.995%

SECTION 4: FIRST AID MEASURES

- Inhalation : Remove to fresh air. If breathing has stopped or is labored, give assisted respirations. Supplemental oxygen may be indicated. If the heart has stopped, trained personnel should begin cardiopulmonary resuscitation immediately. In case of shortness of breath, give oxygen.
- Skin : If frostbite or freezing occur, immediately flush with plenty of lukewarm water (105-115°F; 41-46°C). DO NOT USE HOT WATER. If warm water is not available, gently wrap affected parts in blankets. Get immediate medical attention.
- Eyes : Flush eyes with plenty of water for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get immediate medical attention.
- Ingestion : If swallowed, get medical attention
- Most Important Symptoms/Effects
- Acute : Suffocation, frostbite

SECTION 5: FIRE FIGHTING MEASURES

- Suitable Extinguishing Media : All known extinguishing media can be used.
- Special hazards : Upon exposure to intense heat or flame, cylinder will vent rapidly and or rupture violently. Product is non-flammable and does not support combustion. Move away from container and cool with water from a protected position. Keep containers and surroundings cool with water spray.
- Special protective equipment for fire-fighters : Wear self-contained breathing apparatus for fire fighting if necessary.

SECTION 6: ACCIDENTAL RELEASE MEASURE

Personal precautions, protective equipment and emergency procedures:

Evacuate personnel to safe areas. Wear self-contained breathing apparatus when entering area unless atmosphere is proved to be safe. Monitor oxygen level. Ventilate the area.

- Environmental Precautions : Do not discharge into any place where its accumulation could be dangerous. Prevent further leakage or spillage if safe to do so.
- Methods and material for containment and cleaning up : Stop leak if possible without personal risk. Keep unnecessary people away, isolate hazard area and deny entry. Stay upwind and keep out of low areas. Ventilate closed spaces before entering. Damaged cylinders should be handled only

by specialists. Do not touch or walk through spilled material. If possible, turn leaking containers so that gas escapes rather than liquid.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling	: Avoid breathing dust/fume/gas/mist/vapors/spray. Use only with adequate ventilation
Conditions for Safe Storage,	: Protect from sunlight.
Including any	Store in a well-ventilated place.
Incompatibilities	Store and handle in accordance with all current regulations and standards. Subject to storage regulations: U.S. OSHA 29 CFR 1910.101. Keep separated from incompatible substances.
Incompatible Materials	: Metals, oxidizing materials

SECTION 8: EXPOSURE CONTROL AND PERSONAL PROTECTION

Appropriate engineering controls	: Provide natural or mechanical ventilation to prevent oxygen deficient atmospheres below 19.5% oxygen.
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Individual protection measures, such as personal protective equipment:

Respiratory protection	: Self-contained breathing apparatus (SCBA) or positive pressure airline with mask are to be used in oxygen-deficient atmosphere. Air purifying respirators will not provide protection. Users of breathing apparatus must be trained.
Hand protection	: Wear working gloves when handling gas containers. Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary.
Eye/face protection	: For the gas: Eye protection not required, but recommended. For the liquid: Wear splash resistant safety glasses. Contact lenses should not be worn. Provide an emergency eye wash fountain and quick drench shower in the immediate work area.
Skin and body protection	: For the gas: Protective clothing is not required. For the liquid: Wear appropriate protective, cold insulating clothing.
Special instructions for protection and hygiene	: Ensure adequate ventilation, especially in confined areas.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Physical state	: Compressed gas
Colour	: Colourless.
Odour	: Odourless
Melting point	: -346 °F (-210 °C)
Boiling point	: -321 °F (-196 °C)
Flammability (solid, gas)	: Refer to product classification in Section 2
Vapor pressure	: 760 mmHg @ -196°C
Viscosity	: 0.01787 cp
Solubility in Water	: 0.02 g/l
Relative vapour density	: 0.97 (air = 1) Lighter or similar to air.
Molecular weight	: 28 g/mole
Density	: 1.2506 g/L
Specific Volume	: 13.80 ft ³ /lb (0.8615 m ³ /kg) at 70 °F (21 °C)
Volatility	: 100%
Solvent solubility	: Soluble – liquid ammonia Slightly soluble – alcohol

SECTION 10: STABILITY AND REACTIVITY

Chemical stability	: Stable under normal conditions.
Possibility of	: Will not polymerize
Hazardous Reactions	
Conditions to avoid	: Protect from physical damage and heat. Containers may rupture or explode if exposed to heat.
Hazardous decomposition products	: oxides of nitrogen

SECTION 11: TOXICOLOGICAL INFORMATION

Inhalation Effects	: nausea, vomiting, tingling sensation, suffocation, convulsions, coma, headache, drowsiness, dizziness, loss of coordination, Unconsciousness, fatigue, impairment of judgement, irregular heartbeat
Skin contact	: blisters, frostbite
Eye Contact	: frostbite, blurred vision
Ingestion Effects	: Ingestion is not considered a potential route of exposure.
Product Toxicity Data	
Immediate effects	: Suffocation, frostbite
Specific Target	: Simple Asphyxiant
Organ Toxicity - Single Exposure	

SECTION 12: ECOLOGICAL INFORMATION

No known ecological damage caused by this product.

SECTION 13: DISPOSAL INFORMATION

Waste from residues / unused products : Contact supplier if guidance is required. Return unused product in original cylinder to supplier.
Contaminated packaging : Return cylinder to supplier.

SECTION 14: TRANSPORT INFORMATION

UN Number : UN 1066
UN Proper Shipping Name : NITROGEN, COMPRESSED
Hazard class / Division : 2.2 (Non-flammable gas)



SECTION 15: REGULATORY INFORMATION

- Occupational Safety and Health Act 1994 and Regulations.
- Use and Standards of Exposure of Chemical Hazardous to Health Regulation 2000.
- Occupational Safety and Health (Classification, Labeling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013.

Country	Regulatory list	Notification
USA	TSCA	Included on Inventory.
EU	EINECS	Included on Inventory.
Canada	DSL	Included on Inventory.
Australia	AICS	Included on Inventory.
South Korea	ECL	Included on Inventory.
China	SEPA	Included on Inventory.
Philippines	PICCS	Included on Inventory.
Japan	ENCS	Included on Inventory.

SECTION 16: OTHER INFORMATION

- NFPA Ratings : Health: 0 Fire: 0 Instability: 0 Other: SA
Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate 3 = Serious 4 = Severe
- Training information : Users of breathing apparatus must be trained. The hazard of asphyxiation is often overlooked and must be stressed during operator training. Ensure operators understand the hazards.
- Other information : Before using this product in any new process or experiment, a thorough material compatibility and safety study should be carried out. Ensure adequate air ventilation. Ensure all national/local regulations are observed. Whilst proper care has been taken in the preparation of this document, no liability for injury or damage resulting from its use can be accepted.